

Performance Comparison Between Discrete-time Differentiators for Sensorless PMSM Drives

1st Emmanuel Adamski de Moura

Power Electronics and Control Group - GEPOC

Federal University of Santa Maria

Santa Maria, Brazil

emmanomoura@gmail.com

2nd Pedro Henrique Bulegon Löbner

Power Electronics and Control Group - GEPOC

Federal University of Santa Maria

Santa Maria, Brazil

phlobler@gmail.com

3th Carlos Henrique Barriquello

Study and Development of Electronic Ballasts Group - GEDRE

Federal University of Santa Maria

Santa Maria, Brazil

barriquello@gedre.ufsm.br

Abstract—The efficiency and precision in controlling permanent magnet synchronous machines (PMSMs) are critical for various industrial and automotive applications, especially in electric vehicle traction. This paper presents an in-depth study of discrete-time differentiation techniques for estimating rotor speed and position in PMSMs, which are essential for effective machine operation. Traditional sensor-based methods, while accurate, can be complex, expensive and prone to failures. Therefore, sensorless techniques have been developed to overcome these limitations. Most popular sensorless methods are based on the machine's Back-EMF (BEMF) estimation. In synchronous machines, the BEMF speed is proportional to the rotor speed. In order to observe this variable, a simple approach is the differentiation of the BEMF's electrical angle. Several differentiation methods are compared, including first-order, central difference, state variable filter structure (SVF), super-twisting sliding mode (STSM), and phase-locked loop (PLL). Through detailed mathematical analysis and simulation, the techniques that best mitigate noise and provide accurate estimates, both in transient and steady-state conditions, are identified. Preliminary results suggest that SVF and STSM techniques offer superior performance for the proposed application.

Index Terms—Permanent Magnet Synchronous Machines, Rotor Speed Estimation, Position Estimation, Sensorless Control, Discrete-Time Differentiation Techniques.

I. INTRODUCTION

The efficiency and precision in controlling permanent magnet synchronous machines (PMSMs) are crucial for various industrial and automotive applications, particularly in electric vehicle traction. PMSMs can be classified based on their back electromotive force (EMF) waveform into two main types: sinusoidal and non-sinusoidal [1], [2]. Machines with non-sinusoidal back EMF, known as Brushless DC (BLDC) motors, exhibit a trapezoidal back EMF and are typically driven by three-phase inverters. These BLDC motors are distinguished by their high efficiency, low maintenance requirements, high power density, and compact size, making them ideal for applications where space and performance are critical [3], [4].

Accurate information regarding rotor speed and position is essential for the effective operation of these machines.

Traditionally, such data is obtained through physical sensors. However, the complexity of drive and measurement systems, coupled with the potential for sensor failures, can significantly impact traction control systems [5]. To address these challenges, sensorless techniques have been developed, allowing the estimation of these critical quantities without the need for physical sensors.

Since the 1980s, the literature has explored various techniques to estimate quantities that cannot be directly measured in electric machines [6], [7]. The motivation for developing these techniques lies in the inability to directly measure certain quantities, such as back EMF, and the limitations associated with sensors, including unmodeled dynamics, high costs, and physical space requirements.

In state observer algorithms for sensorless electric motor drive systems, discrete time differentiation techniques play a major role in obtaining the rotor speed. To perform Indirect Field Oriented Control (IFOC), is necessary to obtain both position and speed information. The electric position information can be obtained directly by the BEMF observer. This article focuses on discrete-time differentiation techniques for obtaining rotor speed and position in PMSMs, highlighting their importance and applicability in modern control systems.

There are multiple discrete-time differentiators methods presented in the scientific literature, such as Forward Euler, Backward Euler, central difference or Tustin [8], [9]. However, discrete time differentiators have an inherent noise amplification characteristic. Some methods can cause phase delay or instability. Therefore, the analysis and comparison of different discrete time differentiation techniques can be of great value for the study of non-sinusoidal PMSMs, since the harmonic content present in the BEMF is contained in the electrical position obtained through observers.

Among the existing methods, the following stand out: first-order, central difference, state space filter (SVF), super-twisting sliding mode (STSM), and phase-locked loop (PLL).

Several works have already applied these techniques. In [10]

the low-cost and high-performance of a bearingless induction motor (BIM), a speed sensorless control strategy combined with the improved sliding mode observer (SMO) and phase locked-loop (PLL) is proposed.

In [11], the differences in the requirements for the sliding mode behavior for continuous and discrete-time systems are presented, and the limitations of discrete-time variable structure sliding mode control are discussed. In [12], the paper describes the application of the sliding mode variable structure control (SMVSC) concept for position control of direct current (DC) motors. It presents a mathematical procedure for determining the values of the gains of the variable structure system (VSS) controller and the slope of the switching line.

In [13], a controller based on the super-twisting (ST) algorithm with variable gains is presented, utilizing Lyapunov techniques. This controller was developed to control a wind energy conversion system (WECS) connected to the grid with slip power recovery and electronic control. The controller aims to maximize energy conversion efficiency in the partial load zone, where the wind speed is lower than the turbine's rated speed, thereby reducing mechanical fatigue and mitigating output oscillation. Another example is [14], where the article presents a comprehensive approach to optimizing PMSM control systems by combining advanced observer techniques and sliding mode control strategies. The proposed methods aim to improve tracking accuracy, reduce chattering, and enhance the system's robustness against disturbances.

This paper is organized as follows: the mathematical modeling and control strategies of the PMSM are described in Section II, where the equations governing the motor's behavior in different reference frames are detailed, and the principles of Indirect Field Oriented Control (IFOC) are introduced. Section III presents the design and implementation of the BEMF observer, including the observer model, convergence algorithm, and the derivation of rotor speed and position from the observed BEMF. The discrete-time differentiation techniques used for estimating rotor speed are systematically analyzed in Section IV, covering first-order Euler, central difference, state variable filter (SVF), phase-locked loop (PLL), and super twisting sliding mode (STSM) methods. Finally, conclusions are presented in Section V, summarizing the findings and suggesting directions for future work.

II. PMSM MODEL

The drive system of PMSM is usually composed by a three-phase inverter, aiming to inject electrical currents phase shifted by 120° . The drive system is presented in Figure 1. Indirect Field Oriented Control (IFOC) is applied, allowing independent control of the machine flux and electromagnetic torque by injecting currents into the direct and quadrature axes. The mathematical model of the PMSM is presented in (1), where v_{abc} are the three-phase stator voltages, i_{abc} are the three-phase stator currents, R_s is the stator resistance, L_s is the stator inductance and e_{abc} are the machine's BEMFs.

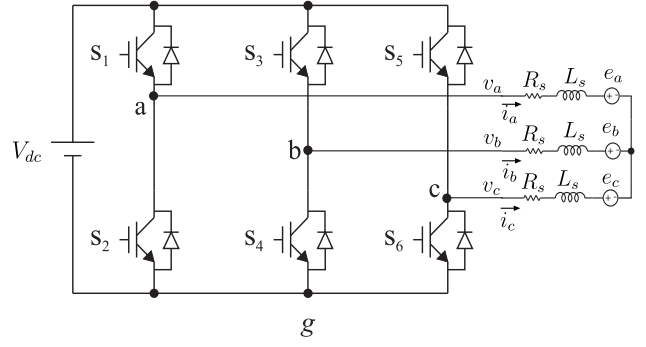


Fig. 1. PMSM drive system.

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} L_s & 0 & 0 \\ 0 & L_s & 0 \\ 0 & 0 & L_s \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} e_a(\theta_r) \\ e_b(\theta_r) \\ e_c(\theta_r) \end{bmatrix} \quad (1)$$

The machine's BEMFs can have sinusoidal or non-sinusoidal behavior, depending on constructive characteristics, like magnet placing [15] and permanent magnet flux linkage behavior. The BEMFs can be obtained by (2), where K_e is the PMSM electric constant, ω_r is the mechanical speed and $e_n(\theta_r)$ is the normalized shape of the BEMF, containing all its harmonic content.

$$e_{abc}(\theta_r) = K_e \omega_r e_{n_{abc}}(\theta_r) \quad (2)$$

The machine model can also be represented in stationary frame, as stated in (3), where $i_{\alpha\beta}$, $v_{\alpha\beta}$ and $e_{\alpha\beta}$ are the stator currents, voltages and the machine BEMF in $\alpha\beta$ coordinates. The stationary reference frame model is particularly useful in BEMF observer techniques.

$$\frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = -\frac{R_s}{L_s} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{1}{L_s} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} - \frac{1}{L_s} \begin{bmatrix} e_\alpha \\ e_\beta \end{bmatrix} \quad (3)$$

To perform the vector control, the synchronous reference frame is applied, in order to utilize linear regulators, like proportional-integral, in the current loop. The PMSM model in synchronous coordinates is given by (4), considering $L_d \approx L_q \approx L_s$, where v_{dq} are the direct and quadrature axes voltage components, i_{dq} are the direct and quadrature axes current components, ω_e is the electric speed and λ_{pm} is the permanent magnet flux.

$$\begin{bmatrix} v_q \\ v_d \end{bmatrix} = \begin{bmatrix} R_s & 0 \\ 0 & R_s \end{bmatrix} \begin{bmatrix} i_q \\ i_d \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} L_s & 0 \\ 0 & L_s \end{bmatrix} \begin{bmatrix} i_q \\ i_d \end{bmatrix} + \omega_e \begin{bmatrix} L_s & 0 \\ 0 & -L_s \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \omega_e \lambda_{pm} \\ 0 \end{bmatrix} \quad (4)$$

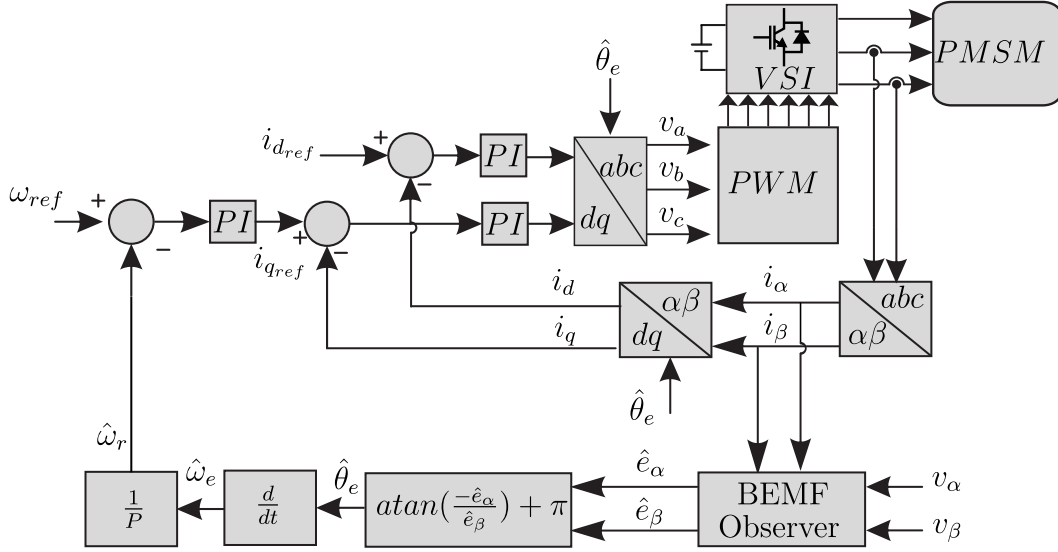


Fig. 2. Sensorless speed control diagram of the PMSM.

The electromagnetic torque T_e generated by the PMSM is given by (5).

$$T_e = \frac{(e_\alpha i_\alpha + e_\beta i_\beta)}{\omega_e} \quad (5)$$

The dynamic behavior of the rotor speed is defined by (6), where B is the viscous friction coefficient and J is the inertia coefficient of the PMSM.

$$\frac{d\omega_r}{dt} = \frac{-B}{J}\omega_r + \frac{1}{J}(T_e - T_l) \quad (6)$$

The sensorless control diagram based on IFOC technique is presented in Figure 2.

III. SPEED AND BEMF OBSERVER

There are several model-based BEMF estimation methods presented in the scientific literature, like Luenberger Observers [16], disturbance observers [17] and Sliding Mode Observers [18]. In general, model-based BEMF observers don't have a good performance in low speeds, converging in medium and high rotor speeds. In [19], a model-based BEMF observer built in stationary reference frame, using linear regulators is presented. This technique is chosen by project simplicity and good dynamic response when compared to the Luenberger observer.

The observation model presented in (7) is based on the measured currents and control loop voltages. The variables $\hat{i}_{\alpha\beta}$ and $\hat{e}_{\alpha\beta}$ are the observed currents and BEMFs in stationary reference frame. The unknowns measurements, $e_{\alpha\beta}$ will converge when the error between observed currents and real currents $\tilde{i}_{\alpha\beta}$ converge to zero.

$$\frac{d}{dt} \begin{bmatrix} \hat{i}_\alpha \\ \hat{i}_\beta \end{bmatrix} = -\frac{R_s}{L_s} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{1}{L_s} \begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} - \frac{1}{L_s} \begin{bmatrix} \hat{e}_\alpha \\ \hat{e}_\beta \end{bmatrix} \quad (7)$$

$$\tilde{i}_{\alpha\beta} = \hat{i}_{\alpha\beta} - i_{\alpha\beta} \quad (8)$$

The convergence algorithm for the BEMFs observation is based on a proportional-integral regulator, as given by (9), where k_1 is the proportional gain and k_2 is the integral gain.

$$\begin{bmatrix} \hat{e}_\alpha \\ \hat{e}_\beta \end{bmatrix} = k_1 \begin{bmatrix} \tilde{i}_\alpha \\ \tilde{i}_\beta \end{bmatrix} + k_2 \int \begin{bmatrix} \tilde{i}_\alpha \\ \tilde{i}_\beta \end{bmatrix} dt \quad (9)$$

In [19] a design strategy of the observer gains is proposed, based on the analysis of the observer transfer function.

$$\begin{aligned} k_1 &= 2\xi\omega_n L_s \\ k_2 &= \omega_n^2 L_s \end{aligned} \quad (10)$$

Where ξ is the damping constant and ω_n is the cut-off frequency.

The BEMF observer is used as an intermediate step to obtain two unknown variables. The BEMF electrical position, applied to the synchronous reference frame transformation to perform the IFOC algorithm and the mechanical speed, needed to perform speed control. The observed electrical position $\hat{\theta}_e$ can be obtained by (11).

$$\hat{\theta}_e = \text{atan}\left(\frac{-\hat{e}_\alpha}{\hat{e}_\beta}\right) + \pi \quad (11)$$

The motor electric speed can be obtained by performing the differentiation of $\hat{\theta}_e$, as shown in (12). The relationship between electrical and mechanical speed is given by (13), where P is pair pole number.

$$\hat{\omega}_e = \frac{d}{dt} \hat{\theta}_e \quad (12)$$

$$\hat{\omega}_r = \frac{1}{P} \hat{\omega}_e \quad (13)$$

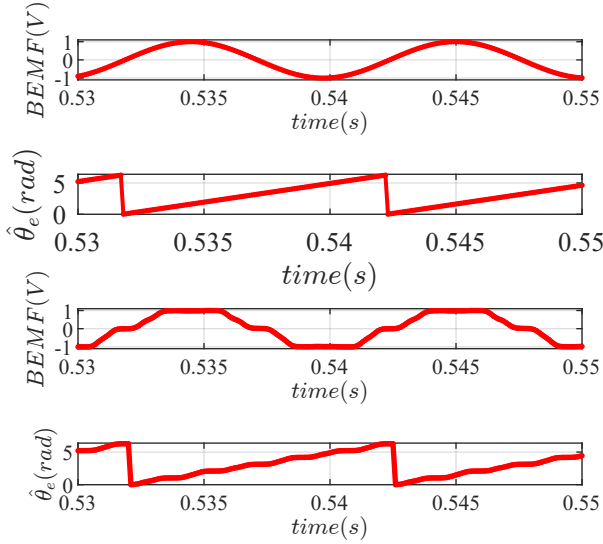


Fig. 3. Results obtained by performing (11) in sinusoidal (first graphic) and non-sinusoidal BEMFs (third graphic.)

When performing (11) in non-sinusoidal PMSMs, the harmonic content presented on the BEMF is reflected to $\hat{\theta}_e$, as outlined in Figure 3. The first graphic shows a typical normalized sinusoidal BEMF, the second graphic represent the $\hat{\theta}_e$ obtained through (11). The third graphic displays a non-sinusoidal normalized BEMF, obtained through experimental setup. The fourth graphic demonstrate the $\hat{\theta}_e$ obtained in this case, highlighting the harmonic content presented in $\hat{\theta}_e$. Therefore, to guarantee an adequate speed observer performance, is necessary to choose a differentiation technique that can reject high frequency noise. Some discrete time differentiation techniques, like first order Euler approximation and central difference, perform a division by the sampling time T_s . This can lead to noise amplification that can compromise the observer effectiveness.

IV. DISCRETE-TIME DIFFERENTIATION TECHNIQUES

The estimation of speed from position data can be performed using discrete-time differentiation techniques. Five methods were selected to determine the rotor speed. The description of these techniques, as well as the results obtained, will be discussed in this chapter.

After the description and mathematical analysis of each applied differentiation technique, it is expected to identify the technique that best rejects noise and presents behavior closest to reality, both in transient and steady-state regimes.

The drive system is simulated via Hardware-in-the-loop (HIL). Therefore, the power source, three-phase inverter and PMSM mathematical model are compiled in real-time in Typhoon HIL 402. The implementation of the sensorless control technique is performed by the STM32-H755ZI microcontroller. The connection scheme between HIL device and microcontroller is presented in Figure 4.

The motor and drive system parameters are presented in Table I. The simulation were performed with $\omega_r^* = 50 \text{ rad/s}$

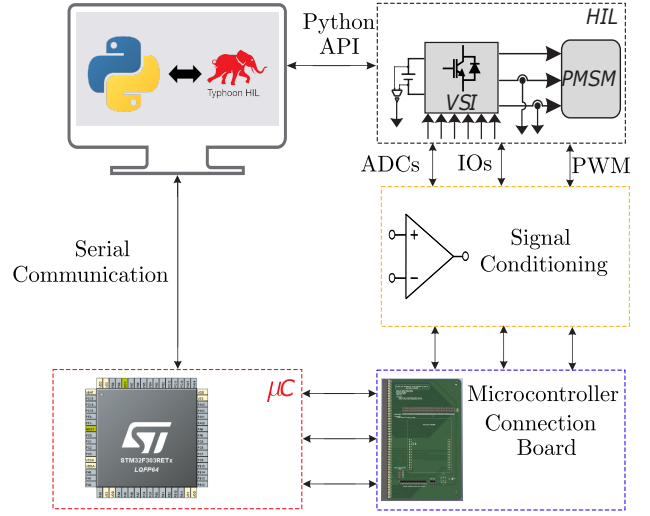


Fig. 4. Connection scheme between HIL device and STM32-H755ZI.

till 0.5 seconds, $\omega_r^* = 100 \text{ rad/s}$ from 0.5 seconds to 1 seconds, $\omega_r^* = 150 \text{ rad/s}$ from 1 second to 1.5 seconds and $\omega_r^* = 200 \text{ rad/s}$ from 1.5 seconds to 2 seconds, with 500 ns time step.

TABLE I
PMSM PARAMETERS

Parameter	Value
V_{cc}	300 V
R_s	3.4 Ω
L_s	0.055 H
J	0.00087 kgm^2
B	0.00058 Nm/s
P	2
K_e	0.377 V/rad/s
T_l	0.2 Nm
f_s	20 kHz
ω_r^*	200 rad/s
ω_n	3141.5926 rad/s
ξ	0.8

A. First-order Euler Differentiation

The First-order Euler Differentiation method is a straightforward technique for estimating the rotor speed. This method utilizes the difference between the current and previous rotor positions, divided by the sampling period, to calculate the speed. The mathematical formulation for this differentiation technique is given by (14).

$$\hat{\omega}_e(k) = \frac{\hat{\theta}_e(k) - \hat{\theta}_e(k-1)}{T_s} \quad (14)$$

where:

- $\hat{\omega}_e(k)$ is the estimated rotor speed at the k -th sample,
- $\hat{\theta}_e(k)$ is the estimated rotor position at the k -th sample,
- $\hat{\theta}_e(k-1)$ is the estimated rotor position at the $(k-1)$ -th sample,
- T_s is the sampling period.

This method approximates the derivative of the rotor position with respect to time, providing an estimate of the rotor speed. It is computationally easy to implement, making it suitable for real-time applications. However, it is sensitive to noise and may not perform well in the presence of significant measurement disturbances. The accuracy of this method depends on the resolution of the rotor position measurements and the chosen sampling period.

The results of this implementation are presented in Figure 5. The graph shows the performance of the First-order Euler Differentiation method, highlighting its response during transient and steady-state conditions.

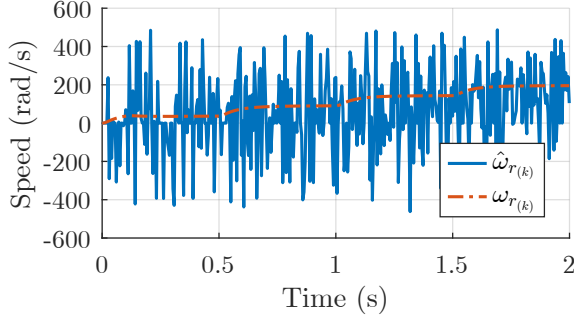


Fig. 5. Comparison between the estimated rotor speed ($\hat{\omega}_e(k)$) and the actual rotor speed ($\omega_e(k)$) over time.

B. Central Differentiation

The Central Differentiation method is another technique for estimating the rotor speed. This method utilizes the difference between the rotor positions two samples apart, divided by twice the sampling period, to calculate the speed. The mathematical formulation for this differentiation technique is given by (15).

$$\hat{\omega}_e(k) = \frac{\hat{\theta}_e(k) - \hat{\theta}_e(k-2)}{2T_s} \quad (15)$$

where:

- $\hat{\omega}_e(k)$ is the estimated rotor speed at the k -th sample,
- $\hat{\theta}_e(k)$ is the estimated rotor position at the k -th sample,
- $\hat{\theta}_e(k-2)$ is the estimated rotor position at the $(k-2)$ -th sample,
- $2T_s$ is twice the sampling period.

This method provides a more accurate estimate of the rotor speed compared to the First-order Euler Differentiation, as it reduces the impact of high-frequency noise by averaging over a longer period. However, it may introduce a slight delay in the speed estimation.

The results of this implementation are presented in Figure 6. The graph shows the performance of the Central Differentiation method, illustrating its ability to smooth out high-frequency noise while tracking the rotor speed.

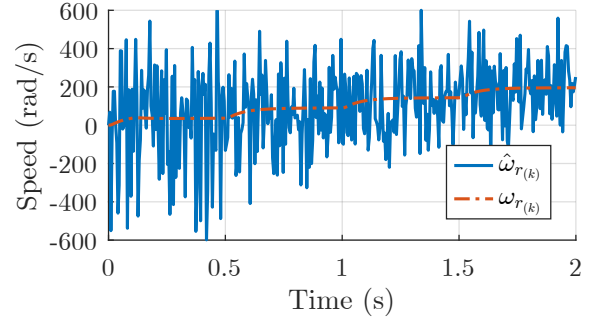


Fig. 6. Comparison between the estimated rotor speed ($\hat{\omega}_e(k)$) and the actual rotor speed ($\omega_e(k)$) over time.

C. State Variable Filter Differentiation

The State Variable Filter (SVF) Differentiation method is an advanced technique for estimating the rotor speed. This method uses a state variable filter to smooth the rotor position signal and reduce noise effects. The mathematical formulation for this differentiation technique is given by:

$$\hat{\theta}_{svf}(k) = \hat{\theta}_{svf}(k-1) + T_s \hat{\omega}_e(k-1) \quad (16)$$

$$\hat{\omega}_e(k) = -T_s \omega_c^2 \hat{\theta}_{svf}(k-1) + (1 - T_s \omega_c^2) \hat{\omega}_e(k-1) + T_s \omega_c^2 \hat{\theta}_e(k) \quad (17)$$

where:

- $\hat{\theta}_{svf}(k)$ is the filtered rotor position at the k -th sample,
- $\hat{\omega}_e(k)$ is the estimated rotor speed at the k -th sample,
- $\hat{\omega}_e(k-1)$ is the estimated rotor speed at the $(k-1)$ -th sample,
- $\hat{\theta}_e(k)$ is the estimated rotor position at the k -th sample,
- $\hat{\theta}_e(k-1)$ is the estimated rotor position at the $(k-1)$ -th sample,
- T_s is the sampling period,
- ω_c is the cutoff frequency of the filter.

This method is beneficial for reducing high-frequency noise and providing a smoother speed estimation. However, it may introduce a slight lag due to the filtering process, depending on the filter design.

The results of this implementation are presented in Figure 7, using a cutoff frequency $\omega_c = 314.1592 \text{ rad/s}$. The graph shows the performance of the State Variable Filter Differentiation method, demonstrating its effectiveness in reducing noise and providing a smooth speed estimate.

D. Phase-Locked Loop Differentiation

The Phase-Locked Loop (PLL) Differentiation method is a sophisticated technique for estimating the rotor speed. This method uses a phase-locked loop to track the rotor position and compute the speed. The mathematical formulation for this differentiation technique is given by:

$$\tilde{\theta}_e(k) = \hat{\theta}_e(k) - \theta_{pll}(k) \quad (18)$$

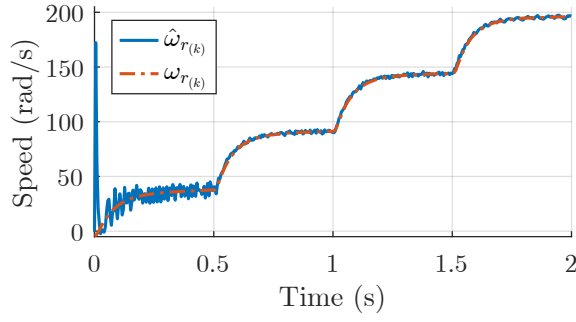


Fig. 7. Comparison between the estimated rotor speed ($\hat{\omega}_e(k)$) and the actual rotor speed ($\omega_e(k)$) over time.

$$\hat{\omega}_e(k) = \hat{\omega}_e(k-1) + (K_I + K_P) \sin(\tilde{\theta}_e(k)) - K_P \sin(\tilde{\theta}_e(k-1)) \quad (19)$$

$$\theta_{pll}(k) = \theta_{pll}(k-1) + T_s \hat{\omega}_e(k-1) \quad (20)$$

The PI regulator gains in continuous-time domain k_p and k_i can be determined by analyzing the closed loop transfer function [19]. The discrete gains K_P and K_I are obtained by Tustin discretization.

$$\begin{aligned} k_p &= 2\xi_{pll}\omega_c \\ k_i &= \omega_c^2 \\ K_P &= \frac{k_p - k_i}{2T_s} \\ K_I &= k_i T_s \end{aligned} \quad (21)$$

where:

- $\tilde{\theta}_e(k)$ is the phase error at the k -th sample,
- $\hat{\theta}_e(k)$ is the estimated rotor position at the k -th sample,
- $\theta_{pll}(k)$ is the PLL's estimated rotor position at the k -th sample,
- $\hat{\omega}_e(k)$ is the estimated rotor speed at the k -th sample,
- $\hat{\omega}_e(k-1)$ is the estimated rotor speed at the $(k-1)$ -th sample,
- T_s is the sampling period,
- K_P and K_I are the proportional and integral gains of the PLL controller, respectively,
- k_p and k_i are the loop filter constants,
- ξ_{pll} is the damping factor,
- ω_c is the cutoff frequency.

This method provides accurate speed estimation by continuously adjusting the estimated position to minimize the phase error. However, it requires careful tuning of the PLL parameters to achieve optimal performance.

The results of this implementation are presented in Figure 8, using $\omega_c = 81.6814 \text{ rad/s}$ and $\xi_{pll} = 0.8$. The graph shows the performance of the Phase-Locked Loop Differentiation method, demonstrating its effectiveness in tracking the rotor speed accurately while minimizing phase error.

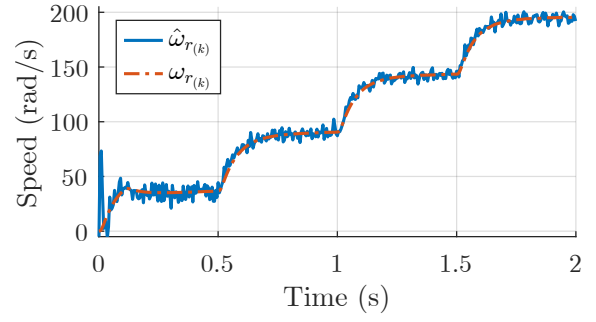


Fig. 8. Comparison between the estimated rotor speed ($\hat{\omega}_e(k)$) and the actual rotor speed ($\omega_e(k)$) over time.

E. Super Twisting Sliding Mode Differentiation

The Super Twisting Sliding Mode (STSM) Differentiation method is a robust technique for estimating the rotor speed, particularly effective in the presence of noise and disturbances. This method uses a second order sliding mode algorithm to track the rotor position and compute the speed. The mathematical formulation for this differentiation technique is given by:

$$\theta_{st}(k) = \theta_{st}(k-1) + T_s \hat{\omega}_e(k-1) \quad (22)$$

$$\sigma(k) = \theta_{st}(k) - \hat{\theta}_e(k) \quad (23)$$

$$u_i(k) = u_i(k-1) - k_2 T_s \text{sign}(\sigma(k-1)) \quad (24)$$

$$\hat{\omega}_e(k) = -k_1 |\sigma(k)|^{\frac{1}{2}} \text{sign}(\sigma(k)) + u_i(k) \quad (25)$$

where:

- $\theta_{st}(k)$ is the estimated rotor position at the k -th sample,
- $\hat{\omega}_e(k)$ is the estimated rotor speed at the k -th sample,
- $\hat{\omega}_e(k-1)$ is the estimated rotor speed at the $(k-1)$ -th sample,
- $\sigma(k)$ is the sliding surface at the k -th sample,
- $u_i(k)$ is the super twisting input at the k -th sample,
- k_1 and k_2 are the sliding mode gains,
- $\text{sign}(\cdot)$ denotes the signal function,
- T_s is the sampling period.

This method is highly effective in rejecting high-frequency noise and provides a robust estimate of the rotor speed even in the presence of disturbances. However, it requires careful tuning of the sliding mode gains to achieve optimal performance.

The results of this implementation are presented in Figure 9, considering $k_1 = 100$ and $k_2 = 170$. The graphic shows the performance of the Super Twisting Sliding Mode Differentiation method, highlighting its robustness in noise rejection and accurate speed estimation at higher speeds, while also illustrating the instability and noise sensitivity at low speeds.

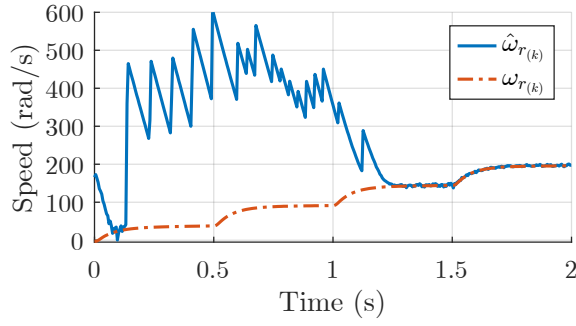


Fig. 9. Comparison between the estimated rotor speed ($\hat{\omega}_e(k)$) and the actual rotor speed ($\omega_e(k)$) over time. Note the instability at low speeds and the improved performance at higher speeds.

The use of HIL simulation allows the evaluation of the computational burden of each proposed differentiation techniques. The results obtained are presented in Table II. It can be noticed that First Order Euler approximation and Central Difference technique presents the lowest computational cost, however both shows inadequate differentiation results, amplifying noises contained in the estimated position. The larger computational burden is achieved by the Phase Lock Loop technique. The State Variable Filter presents a good trade-off between computational burden and speed tracking error.

TABLE II
COMPUTATIONAL BURDEN OF THE EVALUATED TECHNIQUES

Differentiation Technique	Computational Burden
First Order	1.10^{-6} s
Central Differentiation	1.10^{-6} s
State Variable Filter	2.10^{-6} s
Phase Lock Loop	$4.5.10^{-6}$ s
Super Twisting Sliding Mode	$2.5.10^{-6}$ s

V. CONCLUSION

This paper presented a comprehensive performance comparison of various discrete-time differentiation techniques for sensorless control of Permanent Magnet Synchronous Machines (PMSMs). The techniques evaluated include First-order Euler, Central Difference, State Variable Filter (SVF), Phase-Locked Loop (PLL), and Super Twisting Sliding Mode (STSM). Each method was analyzed for its ability to estimate rotor speed and position accurately while mitigating noise and ensuring robust operation under transient and steady-state conditions.

The results indicate that while simple methods like the First-order Euler and Central Difference provide low computational burden, they significantly amplify noise, compromising the accuracy of the speed estimation. On the other hand, advanced techniques such as the SVF, PLL and STSM demonstrated superior performance by effectively rejecting high-frequency noise and maintaining accuracy, even in the presence of disturbances. The PLL method, despite its higher computational cost, showed excellent tracking accuracy and robustness, making it suitable for high-performance applications.

Overall, the SVF method emerged as a balanced approach, offering a good trade-off between computational efficiency and noise rejection capabilities. This makes it highly applicable for real-time sensorless control of PMSMs in electric vehicle traction and other industrial applications.

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